

Simulated Entanglement-Inspired Communication for Federated Quantum Machine Learning Systems

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Abstract

Federated learning enables distributed machine learning without centralized data sharing, offering potential advantages in privacy preservation and scalability. However, federated systems continue to face challenges related to communication overhead, synchronization, non-IID data distributions, and robustness under unstable network conditions — challenges for which no universally satisfactory solution yet exists.

This project proposes a preliminary investigation into a simulated entanglement-inspired communication layer for Federated Quantum Machine Learning (FQML) environments. Rather than relying on physical quantum hardware or genuine quantum entanglement, the project explores whether computational abstractions loosely inspired by entanglement-like correlations between distributed agents might offer any practical benefit. Specifically, it asks whether correlated communication and synchronized probabilistic coordination mechanisms could plausibly improve convergence stability, communication efficiency, or robustness in certain federated settings — and under what conditions, if any, such benefits might materialize.

The internship will focus on the design, implementation, and preliminary evaluation of a prototype federated learning framework incorporating candidate simulated entanglement-inspired coordination strategies. Comparative experiments against standard federated averaging (FedAvg) baselines will be conducted using classical simulation tools. Results may be inconclusive or mixed, and that outcome would itself constitute a valuable finding.

Motivation and Background

Federated learning systems typically require repeated communication between distributed clients and a central aggregator [1, 2]. Communication inefficiencies tend to become more significant in edge AI systems, IoT networks, intermittently connected devices, distributed sensing infrastructures, and quantum machine learning environments [3, 4].

Quantum-inspired approaches have recently attracted attention as potential alternatives for addressing optimization and coordination problems in distributed systems [5, 6]. In particular,

entanglement-inspired correlations offer a conceptual framework for thinking about synchronized probabilistic behavior among distributed agents without continuous communication [7]. Whether this conceptual framing translates into meaningful practical improvements in classical federated systems remains an open and largely unexplored question.

This project tentatively investigates whether simulated entanglement-inspired coordination might:

- reduce communication overhead in certain configurations,
- improve synchronization under specific conditions,
- contribute to training stability under noisy settings,
- or marginally enhance robustness in non-IID federated learning scenarios.

These are hypotheses to be tested rather than anticipated outcomes. The project makes no claim of quantum advantage or physical quantum entanglement. It instead explores a computationally inspired abstraction whose potential utility — and whose limitations — remain to be understood [8, 9].

The present proposal builds naturally on prior work conducted within the group, including the Quantum Federated Learning framework for financial fraud detection developed by Douros and Dalabékis (E.M.P., 2025) [10], which demonstrated the feasibility of hybrid quantum–classical federated architectures in realistic settings. The current project extends this line of inquiry by asking whether the communication layer itself can be enriched with quantum-inspired coordination mechanisms.

Research Objectives

The main objectives are:

1. **Design and critically assess** candidate simulated entanglement-inspired communication mechanisms for federated learning systems.
2. **Implement a prototype** federated learning environment including:
 - a standard FedAvg baseline for comparison [1],
 - and one or more entanglement-inspired communication extensions.
3. **Evaluate**, in a preliminary and exploratory fashion, the possible impact of correlated communication strategies on:
 - model convergence,
 - communication rounds and bandwidth utilization,
 - robustness to noisy or dropped clients,
 - and non-IID data distributions [11, 12].
4. **Reflect** on whether and how the framework might be meaningfully extended toward Federated Quantum Machine Learning scenarios, identifying both promising directions and potential dead ends [4, 13].

Proposed Methodology

The project will proceed through the following stages. Scope adjustments may be necessary depending on what proves feasible within the internship period.

1. Literature Exploration

A focused review will cover:

- Federated Learning fundamentals and known limitations [1, 2, 3],
- Federated Quantum Machine Learning (emergent literature) [4, 13, 14],
- Quantum-inspired distributed optimization and entanglement-based cloud synchronization (e.g., QERA protocols) [5, 6, 24],
- Communication-efficient federated systems [15, 16],
- Correlated randomness and synchronization methods in distributed settings [17, 18].

The review is expected to surface the degree to which similar ideas have already been attempted and what — if anything — distinguishes the present approach.

2. Baseline Federated Learning Environment

A standard federated learning architecture will be implemented using Python-based frameworks such as PyTorch, Flower, TensorFlow Federated, or PennyLane/Qiskit integrations where appropriate [19, 20]. The baseline will include multiple simulated clients, centralized aggregation, non-IID dataset partitioning, and basic communication metrics monitoring [11]. Getting this baseline right is a prerequisite before any entanglement-inspired extensions can be meaningfully evaluated. Michail and Fotios will share responsibility for the baseline implementation, with task allocation to be determined in the opening week in consultation with Epameinondas Douros.

3. Simulated Entanglement Layer

The project will introduce a communication abstraction loosely inspired by entanglement-like correlations [7, 8]. The precise design will likely evolve during the first few weeks as the literature review informs which approaches are most tractable and most distinct from existing techniques. Candidate implementations may include:

- correlated pseudo-random synchronization across clients [17],
- shared latent coordination vectors inspired by multiparty entanglement (e.g., GHZ states) to dynamically link client updates [21, 24],
- teleportation-inspired probabilistic weight updates designed to minimize classical payload transmission [24],
- or lightweight probabilistic coupling between distributed client updates [18].

The design of the coordination mechanism will actively draw architectural inspiration from recent advancements in distributed cloud synchronization—such as Quantum Entanglement-Based Replication Algorithms (QERA) [24]—adapting these multiparty synchronization concepts into

classical shared latent spaces for machine learning. The conceptual objective is to partially emulate nonlocal coordination among agents while remaining executable on classical hardware. It is acknowledged that the analogy to quantum entanglement is metaphorical and that the mechanism may ultimately show limited or negligible benefit — which would still represent a useful empirical result [9].

4. Experimental Evaluation

The proposed method will be compared against standard federated baselines [1, 2] using metrics such as training accuracy and loss curves, convergence speed, communication cost (rounds and total bytes transferred), robustness to delayed or dropped clients, and stability under noisy conditions [12, 15].

Experiments may include simulated "decoherence"-like scenarios represented by packet loss or communication delays to measure the fidelity of the shared state under noise [24], alongside stale gradients, or adversarial noise injection [22, 23]. Results will be reported honestly including null or negative findings, which are considered informative.

Anticipated Outcomes

Rather than promising specific results, the project anticipates producing:

- A working prototype of a candidate entanglement-inspired federated communication layer,
- Preliminary comparative experimental results (whether positive, mixed, or negative),
- A reproducible simulation environment suitable for follow-up work,
- Visualizations and performance analysis with appropriate caveats,
- A short technical report documenting findings, limitations, and open questions.

The primary value of this internship lies in the rigorous exploration of an underexplored idea, not in confirming a predetermined conclusion.

Innovation and Relevance

This project is of potential interest because it bridges conceptual frameworks from quantum-inspired computing and federated AI in a way that remains underexplored in the literature [4, 6, 13], treats entanglement as a communication metaphor rather than a physical claim — which keeps the scope tractable [8, 9] — addresses genuine open problems in synchronization and robustness in distributed AI systems [3, 15], and produces a framework that could serve as a foundation for future educational or research extensions regardless of whether the core hypothesis is confirmed.

The work aligns with emerging research directions in quantum-inspired machine learning [5, 6], distributed intelligence, edge AI [3], and federated quantum systems [4, 13, 14]. It is, however, early-stage exploratory work, and the significance of any findings will depend on experimental outcomes.

Deliverables

By the end of the internship, the student team aims to provide:

1. Source code repository with documentation,
2. Experimental evaluation results with honest analysis,
3. Final technical report (4–6 pages),
4. Presentation slides,
5. Prototype demonstration (scope subject to feasibility).

Timeline

Period	Activities
Week 1 (1–5 July)	Project onboarding; team task allocation; definition of scope and initial architecture
Week 2 (6–12 July)	Literature review; selection of frameworks and datasets
Week 3 (13–19 July)	Design of candidate simulated entanglement protocol; system architecture refinement
Week 4 (20–26 July)	Implementation of baseline federated learning system
Week 5 (27 July – 2 August)	Implementation of entanglement-inspired communication layer

Week 6 (3–9 August)	Experimental evaluation and benchmarking
Week 7 (10–13 August)	Analysis, visualization, and interpretation of results
Final Days (14–15 August)	Final report preparation; presentation and demonstration

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